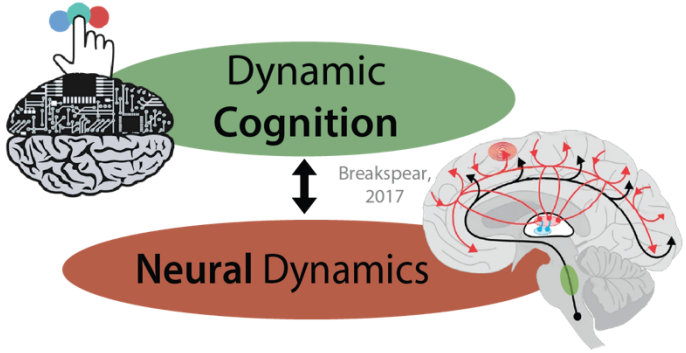
OSCN Pitch – DORO

Julian Kosciessa

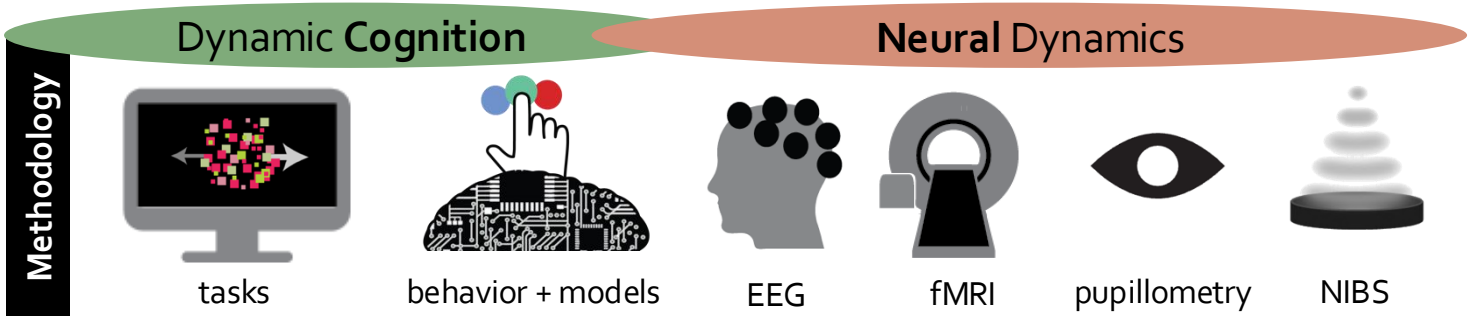
Research Fellow, Donders Centre for Cognition

(feat. slides from Margely Cornelissen)



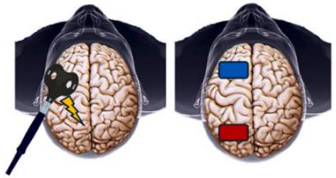
What principles make **computations** flexible?

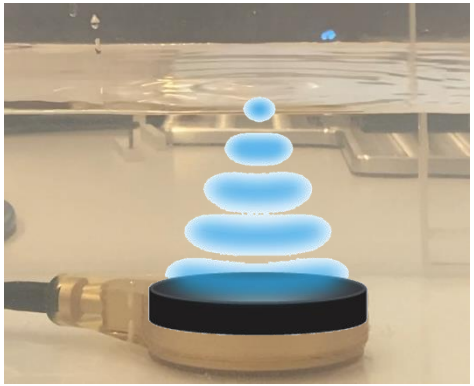
How are they implemented in **brain mechanisms**?



Non-**I**nvasive **B**rain **S**timulation **challenge**:

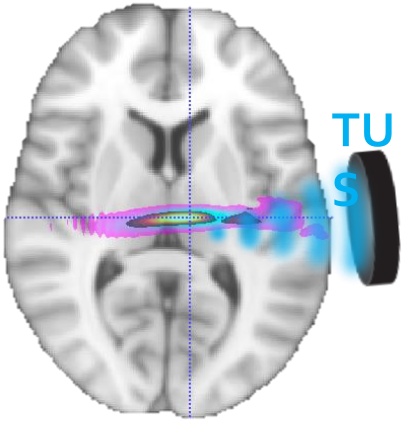
Neuromodulation centers (e.g., thalamus) are **deep, small,** and **differentiated.**





Transcranial **ultrasound** stimulation (TUS) promises neuromodulation that is

spatially **focal & flexible,** temporally **extended,** and offers **deep reach**.



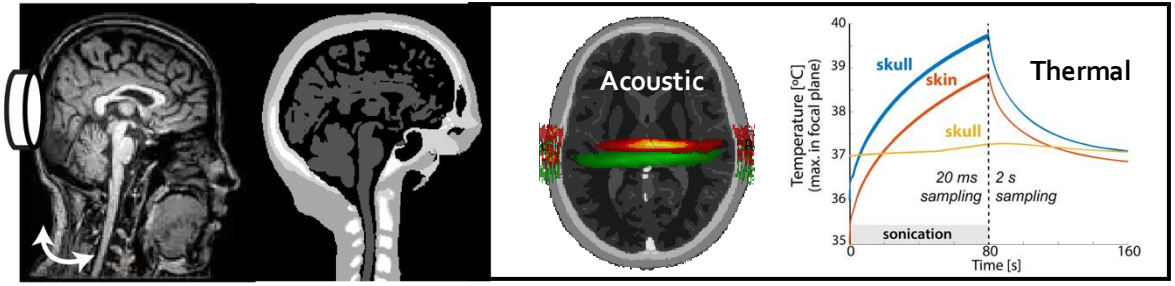
Verhagen et al., 2019, eLife | Folloni, 2022, Science
Nakajima et al., 2022, Cell Reports | Murphy & Fouragnan, 2024, PLOS Biol.



The first-ever **Donders Open Research Output (DORO) Award** recognised outstanding contributions to open science. With this prize, the institute highlights research output that is openly shared, FAIR, and genuinely useful to the wider community, from software to data to tools that help others move their work forward.

This year's inaugural award was presented to Margely Cornelissen and Julian Kosciessa for maintaining open research software for transcranial ultrasound stimulation. The personalised simulation software PRESTUS and open-source hardware interfaces enable rigorous planning, implementation, and reporting of non-invasive ultrasound research. The software, which was co-developed at an earlier stage by former Donders staff members Andrey Chetverikov and Stein Fekkes, has been praised for its high-quality, FAIR-compliant code, robust documentation and practical impact, and offers an open-source alternative to commercial tools. “This openness has even inspired industrial partners to share their own tools,” the committee notes.

PREprocessing & **S**imulations for **T**ranscranial **U**ltrasound **S**timulation



Key features include:

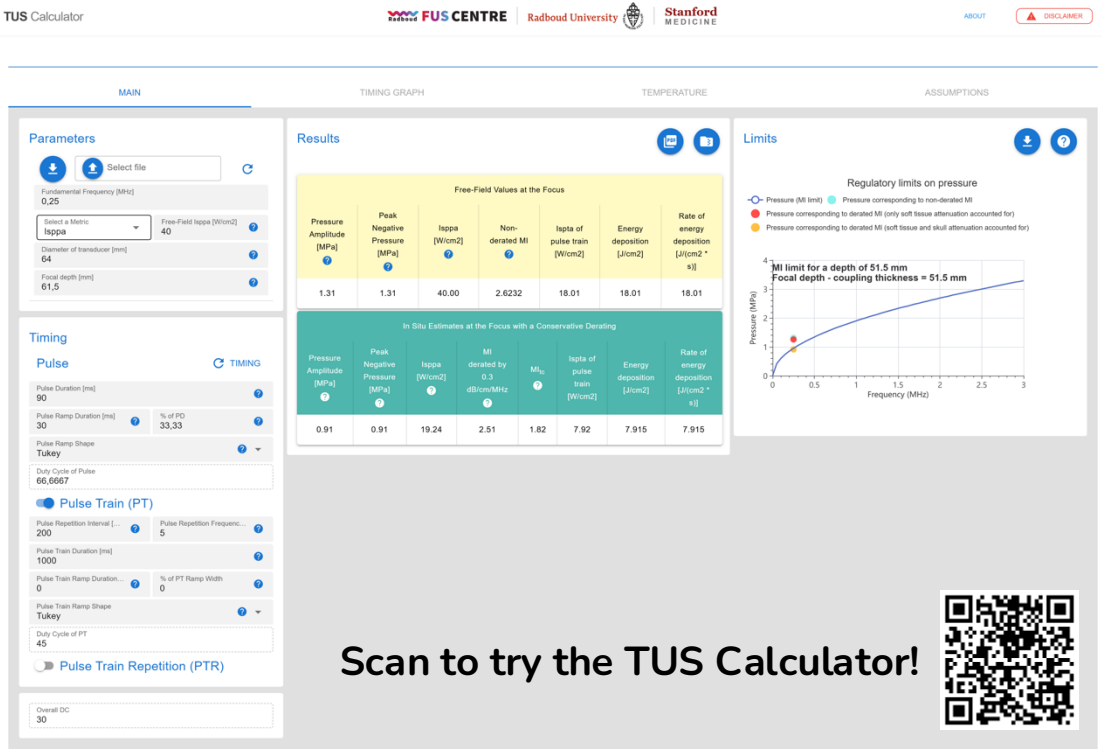
https://donders-institute.github.io/**PRESTUS**/

End-to-end | **Parallel** parameter sweeps
Multi-layer **property mapping**, (p)CT-informed skull support
Heuristic **transducer placement** and localite read-in
Annular **transducer calibration** and matrix transducer support
Flexible protocols (incl. *sync* / *async* / *sequential* multi-transducer mode)
Support for **high-performance computing** (GPU/CPU, C++-optimized)
(In prep.) partial FOVs, additional backends, transducer library

https://github.com/Donders-Institute/**PRESTUS**/

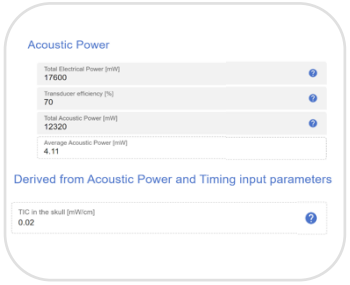
Transducer **Emulation**
Segmentation
pCT generation
Transducer **Placement**
manual | heuristic | PlanTUS | neuronav
Head Preprocessing & **Grid** Setup
Acoustic **Property Mapping**
Source & Sensor Setup
Water Baseline
Acoustic Simulation
Acoustic Analysis
Thermal Simulation
Thermal Analysis
Report Generation

TUS Calculator

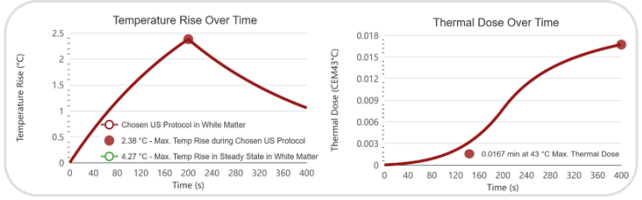


Scan to try the TUS Calculator!

Thermal Index



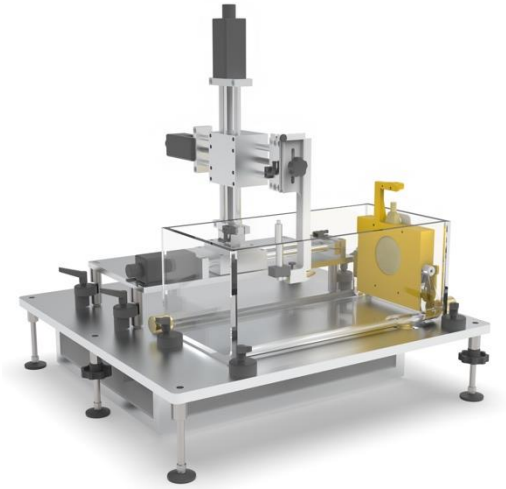
MRI-TUS temperature estimates (beta)



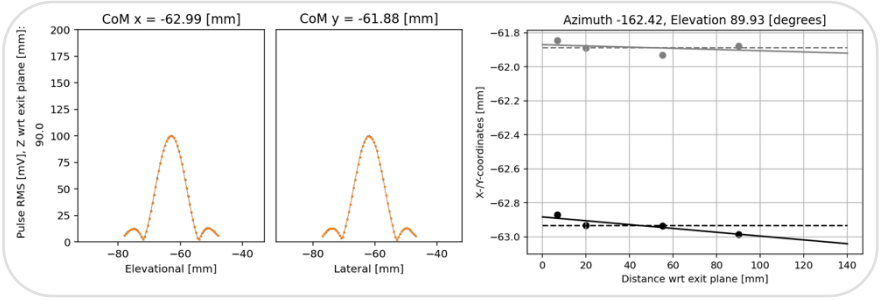
DONDERS INSTITUTE · Radboud FUS CENTRE · VU VRIJE UNIVERSITEIT AMSTERDAM · Stanford MEDICINE School of Medicine · Radboud University · IN DEI NOMINE FELICITER

Radboud FUS Measurement Kit

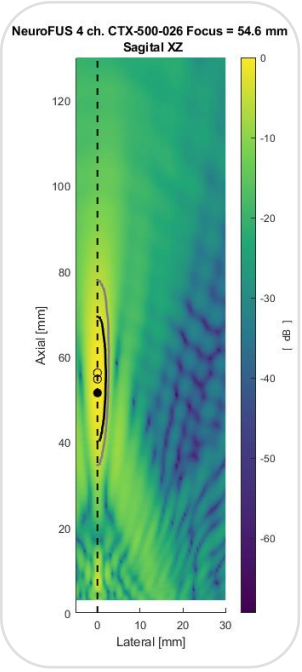
SonoRover One



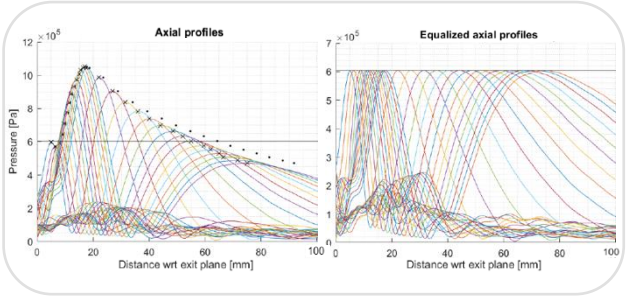
Acoustic alignment



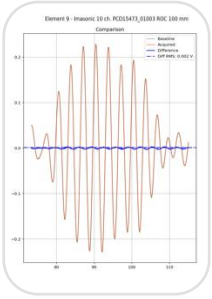
ITRUSST reporting



Calibration



Rapid verification



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Radboud FDS Software

